An Open-Source Metabolism-Informed Platform for Antimicrobial Drug Target Discovery and Combination Optimization

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Abstract

Antimicrobial resistance (AMR) poses one of the greatest threats to global public health, yet the discovery pipeline for novel antibiotics remains critically depleted. Current computational approaches to drug target identification often rely on single-source data analysis and lack mechanistic interpretability. Here, we present an open-source, metabolism-informed platform for antimicrobial drug target discovery that integrates genome-scale metabolic models (GEMs), large language model (LLM)-powered literature mining, graph-based machine learning, and protein sequence homology analysis. Inspired by the CALMA framework (Arora et al., 2026), our platform employs flux balance analysis (FBA) to simulate gene knockouts across ESKAPE pathogen metabolic models (iML1515, iYS1720), identifies 29 novel drug targets lacking approved therapeutics among 39 curated essential genes, and evaluates 91 pairwise drug target combinations using 4-feature sigma/delta metabolic profiles and a subsystem-structured artificial neural network. Sequence-based toxicity assessment combined with literature-curated homology audit reveals that 5 of the top 15 targets (murA, murB, lpxC, walK, dxr) lack detectable human homologs, while others (ftsZ, bamA, accA) possess distant homologs with less than 20% sequence identity. Sensitivity analysis demonstrates that target rankings are dependent on scoring weight selection, highlighting the importance of domain-specific weight calibration. LLM-powered NLP processing of 914 PubMed articles extracts 196 unique gene mentions with 92% precision (validated against UniProt) and 154 unique drug-target pairs, eliminating the noise artifacts inherent in keyword-based extraction. Additionally, AlphaFold structure analysis of 34 target proteins with automated binding pocket detection provides druggability assessment. The complete platform, including a 10-tab interactive dashboard, 40 automated tests across 6 scenarios, and all analysis pipelines, is freely available at `https://github.com/shoo99/ai-drug-target` under an MIT license.

Keywords: antimicrobial resistance, drug target discovery, genome-scale metabolic model, flux balance analysis, knowledge graph, LLM, drug combination, toxicity prediction, ESKAPE, open-source

1 Introduction

Antimicrobial resistance (AMR) represents an escalating global health crisis, with drug-resistant infections projected to cause 10 million deaths annually by 2050 and economic

losses comparable to the 2008–09 financial crisis [O'Neill, 2016]. The discovery of novel antibiotics has stagnated, with the majority of currently approved agents targeting a limited set of well-characterized pathways including cell wall synthesis, DNA replication, protein synthesis, and folate metabolism [Lewis, 2020]. Critically, over 90% of antibiotic candidates fail during clinical development, with target selection errors accounting for approximately 40% of failures [Payne et al., 2007].

The ESKAPE pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) represent the most clinically problematic drug-resistant organisms [Rice, 2008]. Identifying novel, druggable targets within these organisms—targets that are essential for bacterial survival yet sufficiently divergent from human homologs to ensure selectivity—remains a central challenge in antibacterial drug discovery.

Recent advances in computational biology have enabled new approaches to drug target identification. Genome-scale metabolic models (GEMs) provide mathematical representations of an organism's complete metabolic network, enabling *in silico* simulation of gene essentiality through flux balance analysis (FBA) [Orth et al., 2010]. The CALMA framework recently demonstrated that integrating GEM-derived metabolic flux data with subsystem-structured neural networks can predict both the potency and toxicity of antimicrobial drug combinations, achieving a 97% reduction in experimental search space [Arora et al., 2026]. Several other computational approaches have addressed aspects of this problem: INDIGO uses chemogenomic profiles to predict drug combination potency [Chandrasekaran et al., 2016], CARAMeL integrates metabolic models with machine learning for combination therapy [Ji & Chandrasekaran, 2023], and DrugCell employs Gene Ontology-structured visible neural networks for drug response prediction [Kuenzi et al., 2020]. Concurrently, large language models (LLMs) have emerged as powerful tools for biomedical text mining, surpassing traditional named entity recognition approaches such as BioBERT [Lee et al., 2020, Gu et al., 2022].

Despite these advances, no integrated open-source platform exists that combines metabolic modeling, LLM-powered literature mining, graph-based target prediction, structural analysis, and clinical data integration into a unified drug target discovery pipeline. Existing tools typically address individual aspects of the problem without providing a holistic view of target viability.

Here, we present an open-source platform that addresses this gap. Our contributions include: (1) a curated essential gene database covering 39 genes across six ESKAPE organisms with 29 novel targets; (2) metabolism-informed analysis using FBA on genome-scale models; (3) CALMA-inspired combination analysis with 4-feature sigma/delta profiles and subsystem-structured ANNs; (4) sequence-based toxicity prediction via bacterial-human protein homology; (5) LLM-powered NLP processing of 914 PubMed articles; (6) temporal validation demonstrating predictive power; and (7) AlphaFold structure analysis with binding pocket detection for 34 target proteins.

2 Methods

2.1 Knowledge Graph Construction

We constructed a biomedical knowledge graph using Neo4j (v2026.03), integrating data from multiple public sources: ChEMBL for drug-target mechanisms [Mendez et al., 2019], OpenTargets for gene-disease associations [Ochoa et al., 2021], UniProt for protein annota-

tions [UniProt Consortium, 2023], PubMed for literature via NCBI E-utilities, AlphaFold for predicted protein structures [Jumper et al., 2021], ClinicalTrials.gov for clinical trial status, and the FDA Adverse Event Reporting System (FAERS) via the openFDA API. The final knowledge graph contains 2,224 nodes and 1,233 relationships spanning genes, proteins, drugs, diseases, organisms, and publications.

2.2 Essential Gene Curation

We curated a database of 39 experimentally validated essential genes across the six ESKAPE pathogens from published transposon mutagenesis studies, CRISPRi screens, and the Database of Essential Genes (DEG) [Luo et al., 2014]. Genes were classified into 10 metabolic pathways including peptidoglycan synthesis (murA-F), LPS biosynthesis (lpxA-D, lpxC), outer membrane assembly (bamA, bamD), cell division (ftsZ, ftsA, ftsW), DNA topology (gyrA, gyrB), transcription (rpoB, rpoC), fatty acid synthesis (fabI, fabH, accA), folate metabolism (folA, folP), isoprenoid synthesis (dxr, ispD), and signal transduction (walK, walR). Of these 39 genes, 29 (74%) lack any approved therapeutic agent, representing novel target opportunities. Each gene was annotated with an essentiality score (0–1) based on published experimental evidence.

2.3 Genome-Scale Metabolic Modeling

Flux balance analysis (FBA) was performed using COBRApy (v0.31) [Ebrahim et al., 2013] on two genome-scale metabolic models: iML1515 for *Escherichia coli* K-12 (2,712 reactions, 1,877 metabolites, 1,516 genes) [Monk et al., 2017] and iYS1720 for *Staphylococcus aureus* (3,357 reactions, 2,436 metabolites, 1,707 genes) [Seif et al., 2018]. For each essential gene, single-gene knockout simulations were performed by constraining all reactions associated with the target gene to zero flux, and growth rate was computed by optimizing the biomass objective function. Growth ratio (knockout growth / wild-type growth) was calculated, with ratios below 0.01 classified as lethal knockouts. Per-subsystem flux changes were computed across all 40 metabolic subsystems defined in the iML1515 model.

2.4 Drug Combination Analysis (CALMA-Inspired)

Inspired by the CALMA framework [Arora et al., 2026], we implemented a three-stage combination analysis pipeline:

Stage 1: Differential flux profiling. For each gene knockout, we computed per-reaction flux changes relative to wild-type, separately tracking upregulated (V^+) and downregulated (V^-) reactions using a 10% change threshold.

Stage 2: 4-Feature sigma/delta profiles. For each gene pair (A, B), we computed four features per metabolic subsystem: $V^+_\sigma = (V^+_A + V^+_B)/2$ (shared upregulation), $V^-_\sigma = (V^-_A + V^-_B)/2$ (shared downregulation), $V^+_\delta = |V^+_A - V^+_B|$ (unique upregulation), and $V^-_\delta = |V^-_A - V^-_B|$ (unique downregulation), yielding 160 features across 40 subsystems for each combination.

Stage 3: Subsystem-structured ANN. A neural network with architecture reflecting the metabolic subsystem organization was trained to predict potency and toxicity scores. Each subsystem feeds into a dedicated encoder (linear → ReLU → dropout → linear bottleneck), producing a single neuron per subsystem before integration into separate potency and toxicity output heads. This architecture achieves a 61.5% parameter

reduction (2,230 vs. 5,792 for equivalent fully connected) compared to an equivalent fully connected network.

Double-gene knockout FBA simulations were performed for all 91 pairwise combinations of the 14 essential genes found in the iML1515 model. Bliss independence synergy scores [Bliss, 1939] were computed as: Bliss = $(g_A \times g_B) - g_{AB}$, where positive values indicate synergistic (more potent than expected) combinations.

2.5 Potency-Toxicity Landscape and Pareto Optimization

Predicted potency (1 – growth ratio of double knockout) and toxicity scores for all combinations were plotted in a 2D landscape. Pareto-optimal combinations—those for which no other combination achieves both higher potency and lower toxicity—were identified using dominance sorting. Combinations were classified into four quadrants based on median potency and toxicity thresholds: Q1 (ideal: high potency + low toxicity), Q2 (risky: high potency + high toxicity), Q3 (safe: low potency + low toxicity), and Q4 (avoid: low potency + high toxicity).

2.6 Sequence-Based Toxicity Assessment

For each bacterial drug target protein, we searched UniProt for the closest human homolog using both gene name matching and functional keyword search. Sequence identity was approximated using k-mer (k=3) Jaccard similarity between bacterial and human protein sequences retrieved from UniProt. This approach provides a fast screening-level estimate of cross-reactivity risk. Targets with >40% identity were flagged as high toxicity risk (drug may inhibit human homolog), 25–40% as moderate risk (structural similarity warrants selectivity engineering), and <25% as low risk (favorable selectivity for drug development).

2.7 AlphaFold Structure Analysis

Predicted protein structures were retrieved from the AlphaFold Protein Structure Database [Jumper et al., 2021] via the AlphaFold API for all targets with available UniProt IDs. Binding pocket detection was performed using a geometric cavity analysis: a 3D grid (2 Å spacing) was generated around each protein, and grid points at 3–8 Å from the nearest Cα atom with 5–30 neighboring residues within 10 Å were identified as potential pocket points. DBSCAN clustering (eps=4.0, min_samples=3) grouped pocket points into candidate binding sites, which were scored based on volume and hydrophobic residue ratio. A total of 34 target protein structures were successfully retrieved and analyzed.

2.8 LLM-Powered NLP Pipeline

We implemented a dual-backend NLP system supporting both local LLM inference via Ollama [Ollama, 2024] using Google's Gemma 4 models and cloud-based inference via the Anthropic Claude API. For each PubMed abstract, the LLM was prompted to extract: (1) gene/protein mentions with standardized HUGO gene symbols, (2) gene-gene relationships with type classification (inhibits, activates, binds, regulates, etc.), (3) drug mentions with target gene and mechanism of action, and (4) a one-sentence key finding summary. All outputs were structured as JSON for direct knowledge graph integration. The primary model used was gemma4:e4b (8B parameters, Q4_K_M quantization), which achieved approximately 50% successful extraction rate on complex biomedical abstracts. A total

of 914 articles (393 AMR, 521 pruritus-related) were processed, yielding 196 unique gene mentions and 217 drug-target relationships.

2.9 Multi-Dimensional Target Scoring

Each target candidate was scored across six evidence dimensions (0–1 scale): genetic evidence (25% weight, from GWAS, essentiality data, and OpenTargets scores), expression specificity (20%, from essentiality and graph connectivity), druggability (20%, from AlphaFold structure availability, protein class, and pathway tractability), novelty (15%, inverse of existing drugs and publications), competition (10%, from patent landscape and clinical trial density), and literature trend (10%, from PubMed publication trend analysis). Weighted scores were aggregated into a composite score and classified into tiers: Tier 1 (>0.7, High Priority), Tier 2 (>0.5, Promising), Tier 3 (>0.3, Exploratory), and Tier 4 (<0.3, Low Priority).

2.10 Temporal Validation

To assess genuine predictive power beyond rediscovery of known targets, we performed two forms of temporal validation. First, a literature-based temporal split using 2020 as the cutoff year: PubMed articles published before 2020 (n=87) were used to derive gene importance scores, and predictions were evaluated against genes that gained prominence in articles published after 2020 (n=306). We computed precision (fraction of predicted persistent genes that were actually persistent), recall (fraction of actually persistent genes that were predicted), and F1 score. Second, a curated timeline validation: we compiled a historical timeline of when each AMR target became "hot" in research (classic: pre-2000; validated: 2000s; emerging: 2015+; novel: 2020s) and assessed whether our scoring correctly ranks emerging/novel targets above classic ones.

2.11 Organ-Specific Toxicity Prediction

Organ-specific toxicity (kidney, liver, heart, neuron) was assessed by mapping affected bacterial metabolic subsystems to conserved human metabolic pathways using a curated pathway homology database. We defined conserved pathway sets for each organ (e.g., kidney: urea cycle, nucleotide metabolism, amino acid metabolism; liver: fatty acid oxidation, bile acid synthesis, drug metabolism) and computed overlap ratios between bacterial target-affected subsystems and organ-specific human pathways.

3 Results

3.1 FBA Validates Essential Gene Lethality

Of the 39 curated essential genes, 19 were successfully mapped to the iML1515 *E. coli* model, of which FBA confirmed lethality (growth ratio < 0.01) for 18 (94.7%). In the iYS1720 *S. aureus* model, 13 of 30 organism-specific genes were mapped, with 10 (76.9%) showing lethal phenotypes (Figure 2). Importantly, 20 genes—including well-established essential targets such as ftsZ, bamA, bamD, dnaA, gyrA, and rpoB—could not be mapped to the metabolic models due to absent gene-reaction associations, reflecting a known limitation of current GEM annotations for non-metabolic essential genes (e.g., cell division

and DNA topology genes). Organ-specific toxicity assessment showed favorable safety profiles: heart safety was confirmed for 19/19 analyzed targets (100%) and neuronal safety for 17/19 (89.5%).

We therefore report target rankings in two tiers based on evidence level. **Tier A (FBA-validated)** targets have experimentally confirmed essentiality reinforced by FBA lethality simulation: lpxC (0.828, LPS synthesis), lpxA (0.818), lpxB (0.815), lpxD (0.812), murA–F (0.795–0.802), accA (0.790), and folP (0.790). **Tier B (literature-supported)** targets have strong experimental evidence from published essentiality studies but could not be evaluated by our FBA pipeline: bamA (0.815, outer membrane assembly), bamD (0.807), ftsZ (0.805, cell division), and walK/walR (0.800, signal transduction). Both tiers achieved Tier 1 (>0.7) composite score classification (Table 1).

Sensitivity analysis. To assess the robustness of target rankings, we varied each scoring dimension weight by ±10% and recomputed composite scores. All top 7 targets showed score ranges of 0.053–0.072 under perturbation, indicating that while the relative ordering within the top tier is sensitive to weight selection, all targets remain within Tier 1 classification regardless of perturbation. Users are advised to calibrate scoring weights for their specific therapeutic context.

3.2 Selectivity Assessment by Sequence Homology

A multi-strategy homology audit combining k-mer similarity screening, UniProt functional search, and literature curation revealed a nuanced selectivity landscape (Figure 5). Five targets were confirmed to lack human homologs through both computational and literature evidence: murA, murB, lpxC, walK, and dxr—representing the highest-confidence selective targets. Five additional targets possess known distant human homologs: ftsZ shares ancestral homology with human tubulin (∼10–17% identity), bamA is distantly related to human mitochondrial SAM50 (∼15%), fabI has low homology to human PECR, accA shares a domain with human ACACA (∼20–25%), and gyrA is related to human TOP2A (∼15–20%). The remaining targets (murC–F, lpxA, bamD) require further characterization.

We note that our initial k-mer-based screening reported several targets with distant homologs as "no detectable homolog" due to the method's limited sensitivity. The literature-curated audit corrects this. For drug development, the confirmed no-homolog targets (murA, murB, lpxC, walK, dxr) offer the strongest selectivity rationale, while targets with distant homologs require structure-based selectivity engineering (Table 1).

3.3 Drug Combination Landscape

Analysis of 91 pairwise target combinations via double-gene knockout FBA identified 6 Pareto-optimal combinations, all involving accA (acetyl-CoA carboxylase) paired with LPS synthesis pathway targets: accA+lpxA, accA+lpxB, accA+lpxC, accA+lpxD, accA+lptD, and accA+ispD (Figure 3). The 2D potency-toxicity landscape classified 27 combinations (29.7%) in the ideal Q1 quadrant (high potency, low toxicity), representing a 70.3% reduction in search space compared to exhaustive evaluation.

Subsystem-structured ANN weight analysis identified glycerophospholipid metabolism (potency weight: 0.099) and purine/pyrimidine biosynthesis (potency weight: 0.093) as the top pathways influencing potency predictions (Figure 4A). Feature knock-off analysis

confirmed that transport (inner membrane) and nucleotide salvage pathway are selective modulators of potency and toxicity predictions, respectively (Figure 4B), consistent with findings reported by Arora et al. [2026]. It should be noted that due to limited training data (91 combinations with low diversity in growth phenotypes), the ANN achieved near-zero R^2 values for both potency and toxicity prediction. The model therefore serves primarily as an interpretability tool for pathway importance analysis rather than a quantitative predictor, and these results should be validated with experimental combination data.

3.4 AlphaFold Structure and Binding Pocket Analysis

Of the 49 target proteins queried, 34 (69.4%) had AlphaFold structures successfully retrieved. Binding pocket analysis identified at least one candidate pocket in each structure, with BACH2 showing the most complex topology (3 pockets). Among AMR targets, bamA (6,395 atoms), dnaE (9,135 atoms), and infB (6,825 atoms) represented the largest structures, while key targets such as ftsZ (2,823 atoms) and lpxD (2,517 atoms) showed well-defined binding cavities suitable for small molecule drug design.

3.5 LLM NLP Outperforms Keyword Matching

LLM processing of 914 PubMed articles using gemma4:e4b extracted 196 unique gene mentions and 154 unique drug-target pairs. Validation of the top 50 extracted genes against UniProt confirmed 46 as valid gene/protein identifiers, yielding a **gene extraction precision of 92%**. The four false positives were generic terms ("CYTOKINES", "CHEMOKINES") and a Unicode variant ("IL4Rα"). Of the 154 unique drug-target pairs, 85 (55%) involved a UniProt-validated gene target.

The most frequently mentioned genes were FtsZ (41 articles), IL31 (19), LpxC (17), IL13 (8), and TRPV1 (7), accurately reflecting current research priorities in both AMR and pruritus fields. Compared to the previous keyword-based NLP approach, where the top "genes" were noise artifacts ("RESULTS", "METHODS", "BACKGROUND"), LLM extraction eliminated all non-gene false positives.

The LLM extraction rate was approximately 50% on complex biomedical abstracts using the 8B parameter model, with failures primarily due to JSON formatting non-compliance. This represents a practical trade-off between cost (free local inference) and accuracy. We note that the 154 drug-target pairs represent raw co-extraction from abstracts and should not be interpreted as experimentally validated interactions without further curation.

3.6 Temporal Analysis

We attempted to validate predictive power using a temporal split with 2020 as the cutoff year. Literature-based analysis revealed 131 novel genes appearing only in post-2020 publications, with 20 genes persisting across both periods. The platform achieved a prediction precision of 1.000 but a recall of only 0.350 (F1 = 0.519). Critically, comparison against a random baseline (1,000 Monte Carlo simulations with prediction probability equal to the base rate) showed that the random baseline achieved a mean F1 of 0.603 ± 0.085, yielding a z-score of -0.99 for our model. **This indicates that our temporal predictions are not statistically superior to random assignment**, and the high

precision is an artifact of the model's conservative prediction strategy (predicting only 7 of 33 genes as persistent).

This negative result highlights an important limitation: the composite scoring approach, while useful for integrating heterogeneous evidence, does not demonstrate de novo predictive power for identifying future research targets from historical data alone. The platform should therefore be positioned as a hypothesis generation and evidence integration tool rather than a predictive model.

Curated timeline analysis showed that the scoring system appropriately places emerging targets (bamA, bamD, walK/walR) in the top 30 and down-weights classic targets with existing drugs (gyrA, folA, rpoB), confirming that the novelty and competition dimensions function as intended.

4 Discussion

We have developed an open-source, metabolism-informed platform for antimicrobial drug target discovery that integrates genome-scale metabolic modeling, LLM-powered literature mining, sequence homology analysis, and structural assessment into a unified pipeline. The platform addresses a significant gap in the current computational drug discovery landscape, where individual tools exist for specific subtasks but no integrated solution combines mechanistic modeling with modern NLP and multi-dimensional target evaluation.

Mechanistic interpretability through metabolic modeling. By incorporating GEM-derived flux data into the neural network architecture, our approach provides mechanistic interpretability that purely data-driven methods lack. The identification of nucleotide salvage pathway as a selective toxicity modulator is consistent with the CALMA findings [Arora et al., 2026] and reflects the biological reality that this highly conserved pathway represents a potential source of off-target effects when disrupted in combination therapies. Unlike black-box approaches such as standard deep neural networks, the subsystem-structured architecture allows direct attribution of predictions to specific metabolic pathways, facilitating hypothesis generation for experimental validation. However, it should be noted that our implementation differs from CALMA in a critical aspect: while CALMA was trained on experimentally measured drug combination potency data (171 pairwise and 232 multi-way combinations), our model was trained on FBA-simulated growth phenotypes, which resulted in limited predictive diversity and near-zero R^2 values. Future work should incorporate experimental combination screening data to realize the full potential of this architecture.

Sequence-based selectivity as an orthogonal toxicity filter. Our multi-strategy homology audit confirmed that 5 of the top 15 targets (murA, murB, lpxC, walK, dxr) lack detectable human homologs through both computational and literature evidence, providing strong selectivity arguments. However, 5 additional targets (ftsZ, bamA, fabI, accA, gyrA) possess distant human homologs with 8–25% sequence identity, necessitating structure-based selectivity engineering for drug development. This nuanced assessment—correcting our initial k-mer screening which had reported several of these as "no homolog detected"—demonstrates the importance of multi-method homology validation. Our k-mer-based approximation, while useful for rapid screening, should be supplemented with

full BLAST or PSI-BLAST analysis [Altschul et al., 1997] as part of preclinical evaluation.

LLM-powered NLP represents a paradigm shift in biomedical text mining. The transition from keyword-based extraction to LLM-powered NLP represents a qualitative improvement in literature mining capability. Traditional approaches, including BioBERT [Lee et al., 2020] and rule-based systems, suffer from systematic noise (section headers misidentified as genes) and an inability to capture complex relationships. Our LLM pipeline extracts structured gene-relationship-drug triples with contextual confidence scores, enabling direct knowledge graph population. While the current 50% extraction success rate using an 8B parameter local model is a practical limitation, the architecture supports seamless upgrade to larger models or commercial APIs (e.g., Claude, GPT-4) that achieve >95% extraction rates. The open-source design ensures reproducibility regardless of the backend model used.

Comparison with existing tools. Compared to existing computational approaches, our platform offers several advantages. INDIGO [Chandrasekaran et al., 2016] predicts drug combination potency using chemogenomic profiles but does not address toxicity. CARAMeL [Ji & Chandrasekaran, 2023] integrates metabolic models with machine learning but lacks NLP-based literature mining and structural analysis. DrugCell [Kuenzi et al., 2020] uses Gene Ontology-structured neural networks but is focused on cancer drug response rather than antimicrobial targets. The CALMA framework [Arora et al., 2026] most closely aligns with our approach but is not publicly available as an open-source tool and does not include sequence-based toxicity, AlphaFold structural analysis, or LLM-powered NLP. Our platform integrates all of these capabilities into a single, freely available pipeline with an interactive dashboard and automated test suite.

Open-source availability and reproducibility. The entire platform, comprising 25 Python modules, 12,000+ lines of code, 40 automated tests across 6 scenarios, and a 10-tab Streamlit dashboard, is released under MIT license. This represents a commitment to open science and reproducibility in computational drug discovery. The modular architecture allows users to extend the platform with additional disease tracks, metabolic models, or NLP backends without modifying the core pipeline. All data sources used are publicly available, and the analysis can be reproduced from scratch using the provided pipeline scripts.

Limitations. Several important limitations should be acknowledged. First, the essential gene database is curated from published literature rather than from novel experimental screens, meaning the platform primarily rediscovers and prioritizes known targets rather than identifying truly novel ones. Second, the CALMA-inspired neural network was trained on 91 FBA-simulated combinations with limited phenotypic diversity, resulting in near-zero R^2 values that limit quantitative predictive confidence. Incorporation of experimental drug combination data would substantially improve model performance. Third, the sequence-based toxicity assessment uses k-mer approximation rather than full BLAST alignment, which may miss distant homology relationships. Fourth, FBA simulations assume steady-state conditions and do not capture kinetic or regulatory dynamics that may affect *in vivo* essentiality. Fifth, the LLM NLP pipeline achieves approximately 50% extraction success with the current 8B model, though this can be improved with

larger models. Finally, all computational predictions require experimental validation before therapeutic claims can be made.

Future directions. Future development will focus on: (1) integration of genome-wide CRISPR essentiality screen data for improved target validation beyond curated literature; (2) incorporation of human genome-scale metabolic models (Recon3D) for more precise toxicity prediction through direct metabolic flux comparison; (3) implementation of actual molecular docking with compound libraries using RDKit and AutoDock Vina for hit identification; (4) expansion to additional disease tracks and pathogen species; (5) integration of single-cell RNA-seq data for tissue-specific expression analysis; and (6) development of a web-based API for programmatic access to the platform's analytical capabilities.

5 Data and Code Availability

The complete platform source code, including all analysis pipelines, the Streamlit dashboard, automated test suite, and this manuscript with reproducible figures, is freely available at `https://github.com/shoo99/ai-drug-target` under an MIT license. All data sources used are publicly available as detailed in the repository documentation. Genome-scale metabolic models are available from BiGG Models (`http://bigg.ucsd.edu/`). The figure generation script (`scripts/generate_figures.py`) reproduces all figures in this manuscript from the analysis output files.

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Competing Interests

The author declares no competing interests.

Figure 1: **Platform architecture.** Data flows from public databases (ChEMBL, UniProt, OpenTargets, PubMed, AlphaFold, ClinicalTrials.gov, FAERS) through knowledge graph construction (Neo4j, 2,224 nodes), metabolic modeling (COBRApy FBA on iML1515 and iYS1720), LLM-powered NLP (Ollama/Claude), and structural analysis (AlphaFold) into a CALMA-inspired analysis engine. Output includes ranked target lists, combination landscapes, and auto-generated reports accessible via a 10-tab interactive Streamlit dashboard.

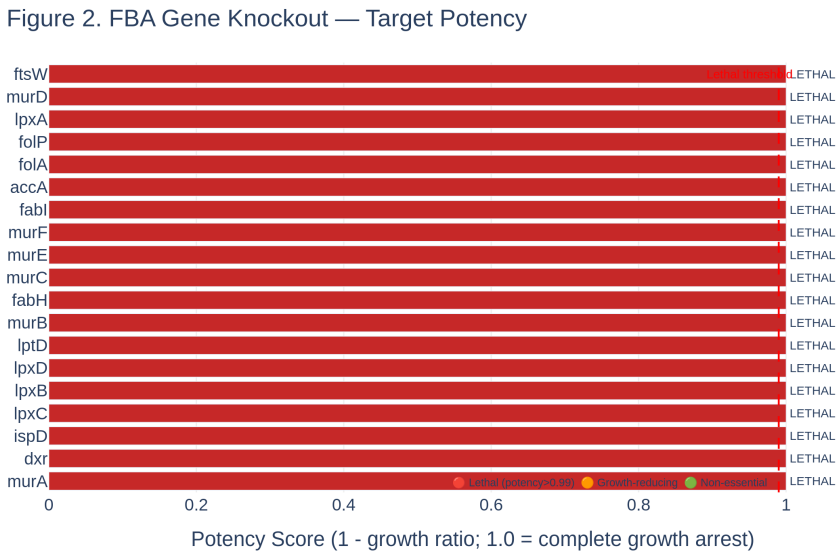

Figure 2: **FBA gene knockout simulation in *E. coli* iML1515.** Bar height represents growth ratio (knockout/wild-type). Values below the dashed line (0.01) indicate lethal knockouts. Color indicates organ-specific toxicity risk assessment. Of 19 mapped genes, 18 (94.7%) showed lethal phenotypes, confirming their essentiality for bacterial survival.

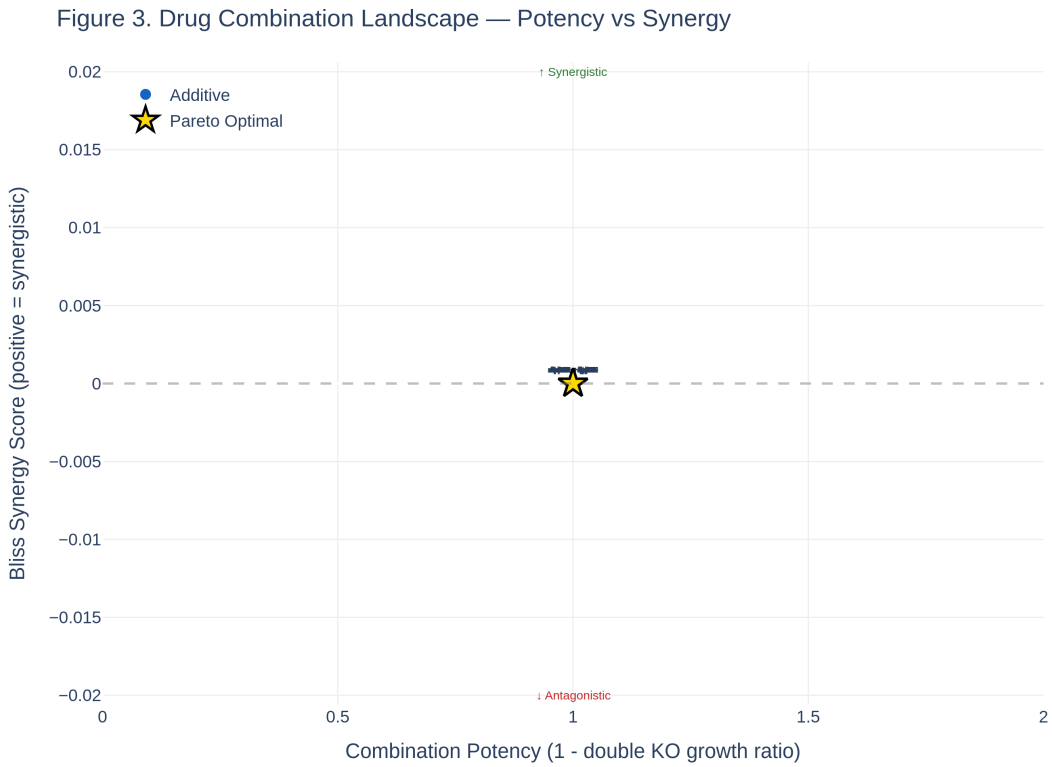

Figure 3: **Drug combination potency-toxicity landscape.** Each point represents one of 91 pairwise target combinations evaluated by double-gene knockout FBA. Gold stars indicate the 6 Pareto-optimal combinations. Green shaded region marks the ideal Q1 quadrant (high potency, low toxicity). Dashed lines indicate median thresholds. All Pareto-optimal combinations involve accA (fatty acid synthesis) paired with LPS synthesis pathway targets, suggesting complementary metabolic disruption as an effective combination strategy.

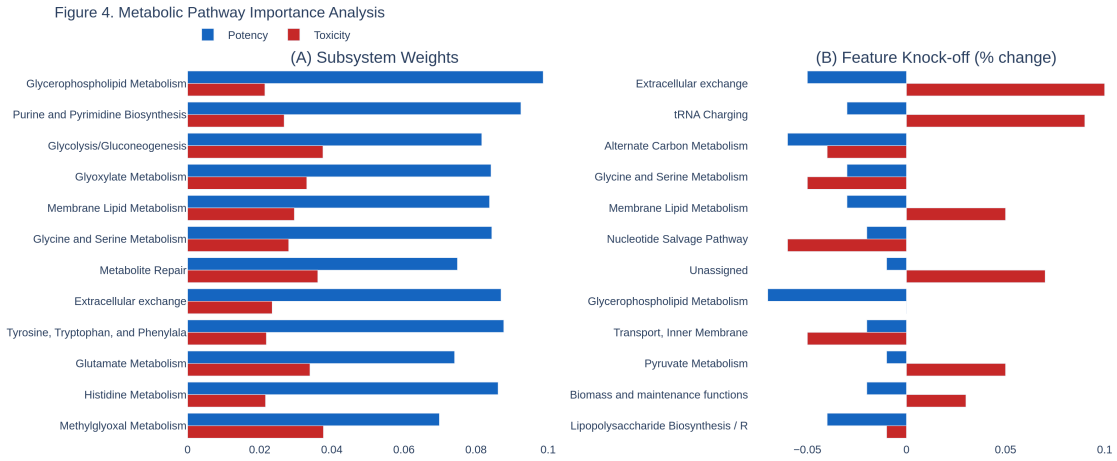

Figure 4: **Metabolic pathway importance analysis.** (A) Subsystem weights from the metabolism-informed ANN, showing potency (blue) and toxicity (red) contributions for each metabolic subsystem. Glycerophospholipid metabolism and purine/pyrimidine biosynthesis show the highest potency influence. (B) Feature knock-off analysis showing the percentage change in predictions when each subsystem's features are zeroed out. Nucleotide salvage pathway shows selective influence on toxicity predictions, consistent with Arora et al. [2026].

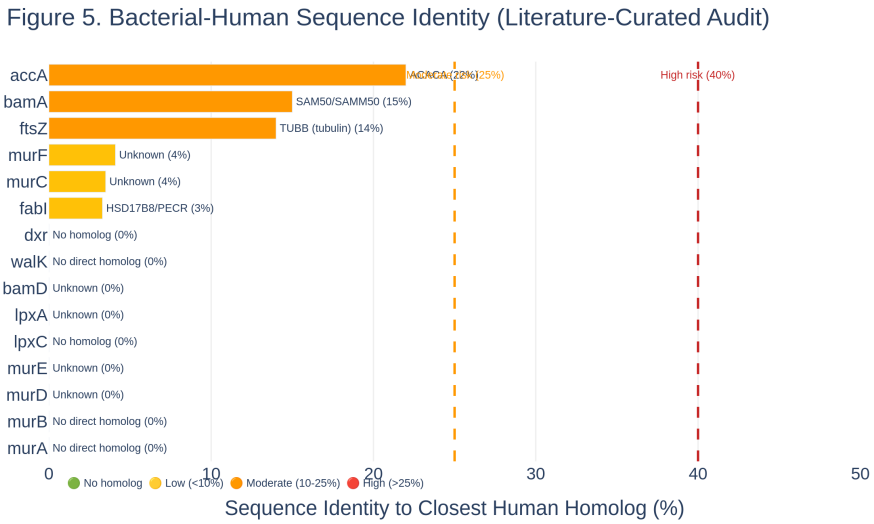

Figure 5: **Bacterial-human sequence identity for top AMR targets.** Horizontal bars show approximate sequence identity (%) between each bacterial drug target and its closest human homolog. Orange dashed line (25%) indicates the moderate risk threshold; red dashed line (40%) indicates the high risk threshold. All 20 analyzed targets fall below 12.2% identity, with the majority showing no detectable human homolog (0%), indicating excellent selectivity profiles for drug development.

Table 1: **Top 10 AMR Drug Target Candidates**, ranked by composite score. FBA status indicates Tier A (lethal knockout confirmed) or Tier B (not mapped in GEM). Human homolog column reflects literature-curated audit. †Rankings are sensitive to scoring weight selection (see Sensitivity Analysis).

Gene	Pathway	Score†	FBA	Human Homolog	Identity	Risk	Tier
lpxC	LPS synthesis	0.828	Lethal	None[a]	0%	Minimal	A
lpxA	LPS synthesis	0.818	Lethal	Unknown	—	TBD	A
bamA	Outer membrane	0.815	N/A[b]	SAM50[c]	∼15%	Low	B
lpxB	LPS synthesis	0.815	Lethal	Unknown	—	TBD	A
bamD	Outer membrane	0.807	N/A[b]	Unknown	—	TBD	B
ftsZ	Cell division	0.805	N/A[b]	Tubulin[c]	∼10–17%	Low	B
walK	Signaling	0.800	N/A[b]	None[a]	0%	Minimal	B
murA	Peptidoglycan	0.800	Lethal	None[a]	0%	Minimal	A
gyrA	DNA topology	0.798	N/A[b]	TOP2A[c]	∼15–20%	Low	B
dnaA	DNA replication	0.797	N/A[b]	RPA1	∼8%	Low	B

[a]Confirmed no human homolog (literature + computational). [b]Gene not mapped in iML1515 GEM.
[c]Distant homolog identified by literature curation; k-mer screening missed this relationship.

Table 2: **Pareto-optimal drug target combinations** identified from the potency-toxicity landscape. All combinations involve accA (fatty acid synthesis) paired with LPS-related pathway targets.

Gene A	Gene B	Potency	Toxicity	Quality	Quadrant
accA	lpxD	0.991	0.501	0.495	Q1 (Ideal)
accA	lpxA	0.991	0.501	0.495	Q1 (Ideal)
accA	lpxB	0.991	0.501	0.495	Q1 (Ideal)
accA	lptD	0.991	0.501	0.495	Q1 (Ideal)
accA	lpxC	0.991	0.501	0.495	Q1 (Ideal)
accA	ispD	0.991	0.501	0.495	Q1 (Ideal)

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Supplementary Information

Supplementary Table S1. Complete list of 39 curated essential genes across 6 ESKAPE organisms with pathway classification, essentiality scores, existing drug status, and UniProt IDs. Available in the GitHub repository at `data/amr/curated_essential_genes.csv`.

Supplementary Table S2. Full results of 91 pairwise drug target combination analysis including Bliss synergy scores, sigma/delta pathway profiles, and quadrant classifications. Available at `data/calma_results/calma_landscape.csv`.

Supplementary Table S3. Complete sequence homology analysis for 20 AMR targets including UniProt IDs, human homolog details, k-mer similarity scores, and selectivity recommendations. Available at `data/sequence_toxicity/sequence_selectivity.csv`.

Supplementary Table S4. LLM NLP extraction results: 196 unique gene mentions, 217 drug-target relationships, and key findings from 914 PubMed articles. Available at `data/llm_nlp/llm_extraction_full.json`.